

A Project Proposal Report on

Smart Money Tracker: A CMD-Based Personal Finance Manager

A Project Report Submitted for the Course

Object-Oriented Programming (OOP)

Bachelor of Science in Computer Science

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ABSTRACT

“Smart Money Tracker” is a simple, command-line based C++ application designed to help a user manage personal finances. The application allows a user to register and log in securely, record in-come, expenses, and custom savings entries with dates, and generate monthly and yearly financial reports. All login credentials and transaction records are stored in CSV files that can be directly opened in Excel for future reference. The system is built using core Object Oriented Program-ming principles – Encapsulation, Inheritance, Polymorphism, and Abstraction – to keep the code organized, reusable, and easy to maintain.

1. INTRODUCTION

Managing personal finances manually or through scattered notes often leads to confusion and loss of track of income, expenses, and savings. Smart Money Tracker is proposed as a lightweight console-based solution that lets a single user record and review their financial activity in an original manner.

1.1 PROBLEM STATEMENT

Many individuals do not maintain a proper record of their daily income and expenses. Without a proper system, it becomes difficult to know how much money is earned, spent, or saved over a month or a year, making financial planning difficult.

1.2 PROJECT OBJECTIVES

- Provide a simple and secure login system for the user.
- Allow the user to record income and expenses with the date.
- Let the user create a custom saving method with amount and date.
- Generate monthly and annual financial reports.
- To store all login and transaction data in a file (CSV, viewable in Excel) for future reference.

1.3 SCOPE AND LIMITATIONS

Scope: • The system is intended for individual, single-user personal finance tracking. • It records income, expenses, and custom savings with date stamps. • It generates simple monthly and yearly summary reports. Limitations: • The system supports only one user account at a time per session. • It is a console (CMD) based application without a graphical interface. • Reports are limited to totals of income, expense, and saving; no graphs are generated.

2. METHODOLOGY

2.1 Tools to be Used •VS Code / Code::Blocks – for writing and editing the C++ code.

•G++ Compiler – for compiling and running the program.

•Git and GitHub – for version control of the source code.

2.2 Technologies to be Used •C++ – the primary programming language used to implement the entire system.

•File Handling (fstream) – used to store login credentials and transaction reports in CSV files.

•Object Oriented Programming – Encapsulation, Inheritance, Polymorphism, and Abstraction are applied throughout the design.

2.1 TOOLS TO BE USED

•C++ – the primary programming language used to implement the entire system.

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•Object Oriented Programming – Encapsulation, Inheritance, Polymorphism, and Abstraction are applied throughout the design.

3. SYSTEM DESIGN

This chapter presents the design of the proposed system through three diagrams: the Class Diagram, the Use Case Diagram, and the Sequence Diagram.

3.1 CLASS DIAGRAM

The class diagram below shows the main classes of the system. Transaction is an abstract base class from which Income, Expense, and Saving inherit. The User class handles login credentials, and MoneyTracker manages the overall operations of the system.

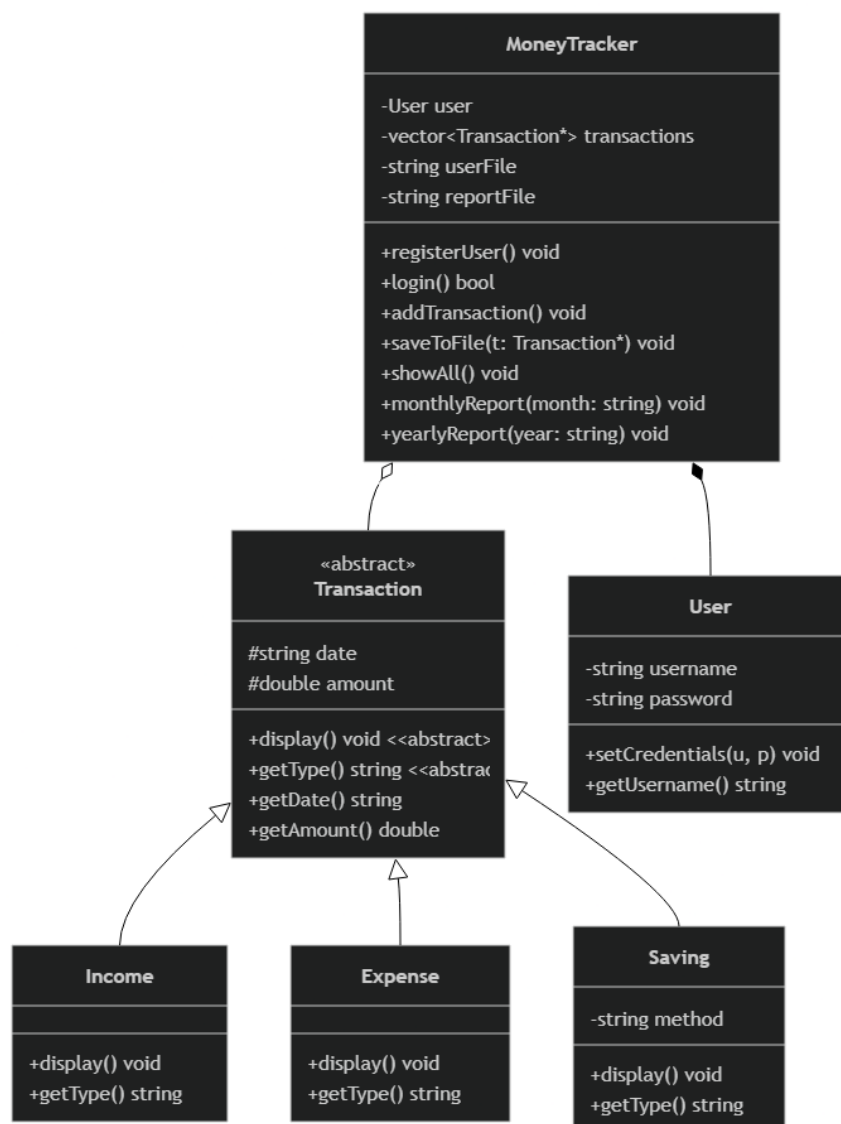


Figure 1: Class Diagram

3.2 USE CASE DIAGRAM

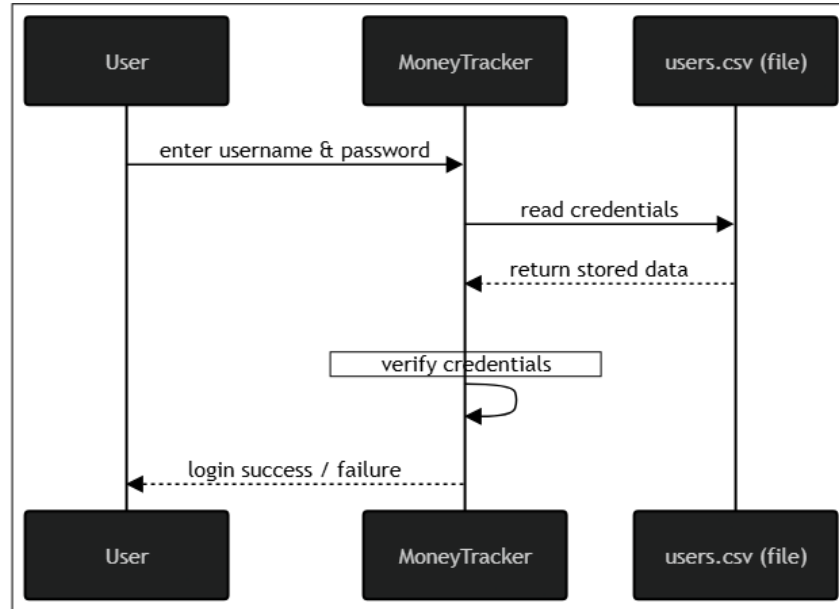
The use case diagram shows the interactions between the User (actor) and the system.



Figure 2: Use Case Diagram

3.3 SEQUENCE DIAGRAM

The sequence diagram below illustrates the flow of the login process between the User, the MoneyTracker system, and the users.csv file.



Test

Figure 3: Sequence Diagram

4. PROPOSED DELIVERABLES

This project will deliver a fully working, console-based Smart Money Tracker application built in C++. The final deliverable will include:

- A registration and login system.
- Functionality to add income, expense, and custom saving entries with date.
- Monthly and yearly report generation.
- CSV files storing login data (users.csv) and transaction records (report.csv), viewable in Excel.

5. CONCLUSION

Smart Money Tracker will help a user manage personal finances in a simple and efficient way through a lightweight C++ console application. The use of Object Oriented Programming concepts – Encapsulation, Inheritance, Polymorphism, and Abstraction – makes the code organized, reusable, and easy to extend in the future.